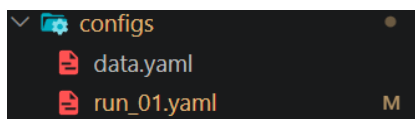


LLM: Model Configuration

Configuration files: `.yaml` extensionga ega bulgan barcha filelar config filelari hisoblanadi.

Configuration ni ichidagi `value` (ex: `vocab_size`, `n_layer`) larga `istalgan son` berishimiz mumkin. yani, bu bizni hoxshimiz.



Vocabulary Size

Vocab_size: model tushuna oladigon jami unique tokenlar soni. Yani model berilgan son gacha vocabulary ochadi va limit tuglanicha suzlarni qabul qiladi. va keyin tuglandan keyin ushandan ishlatadi.

```
vocab_size: 8000
```

`<bos>`: matnni boshiligini bildiruvchi mahsus token (`<eos>` bilan qaramaqarshi)

OOV (out of vocabulary): bizning `vocab_size` imizda `yuq index`ni bersa shu error chiqadi. Yani `IndexError`. Ushanga vocabulary initialization qilayotkanimizda `<unk>` qiymatlarga bittayamta `index berib quyamiz`. Misol `<unk>=10` bersak buladi. yoki `BPE split` usulidan foydalangan holda har bir harfga tegishli indexni topish.

Block-Size (Context-Size)

Block_size: model bir martada uqiy oladigon maximal tokenlar soni.

Agar berilgan block-size dan kup bersa `IndexError` chiqadi.

```
block_size: 256
```

n_embd (Model Dimensional)

n_embd: dense vectorni ichida nechta floating-point sonlari bulishi.

Ex) `n_embd=384` → `[0.4, 0.2, -0.8, .... ~384 ta]` buladi

```
n_embd: 384
```

n_layer

n_layer: transformer blocklarini miqdori. Layerlar soni kup bulgani sari ushancha complex va chuqur urganadi.

```
n_layer: 6
```

n_head

n_head: model berilgan 'n' ta tomondan birdaniga qaraydi. Yani ushancha group bulib ishlaydi

Rule: qoldiq qolmasligi kerak

```
n_head: 6
```

Dropout

Dropout: overfitting ni oldini olish usuli lekin LLM da eskirgan usul. shuning uchun weight decay dan foydalanadi.

```
dropout: 0.1
```

ffn_multi (FFN layer)

ffn_mult: bilimni saqlab qolish uchun model dimensionalga kupaytiriladi. Ex) n_embd=384 bulsa, ffn_mult=4 bulsa, FFN bilim saqlab qolish uchun 1536 vector joy bushatadi.

```
ffn_mult: 4
```

tie_weights=True

tie_weights: embedding bilan LM head ni bitta weight ishlatishini taminlaydi. Bu bulmasa weightlar bekordan duplicate bulib GPU bekordan ortiqcha ishlaydi.

```
tie_weights: true
```